

I Cash Extrapolation Model: From Clover to Settlement

The Clover dataset sheds light on the distribution of cash usage across merchants, which is not directly observable in the settlement dataset. However, Clover merchants are not representative of the composition of the broader economy. We therefore build a model to extrapolate cash usage from Clover to the broader settlement sample. This section details the modeling approach, estimation results, and validation exercises.

I.A Overview

We use a two-part model to extrapolate cash usage from Clover to the broader settlement sample.

1. A logistic regression for the probability that the share of cash transactions at a merchant exceeds 2% (extensive margin).
2. A linear regression for the log of the conditional cash share among merchants with greater than 2% share (intensive margin).

We use a two-stage model because the distribution of cash use in Clover data has a large mass point at zero and is highly skewed among positive values.

An important motivation for using the regression model is that Clover merchants are very different from the merchants in the broader Fiserv settlement sample. They are much smaller and much more likely to be in the restaurant and retail sectors. Thus, we cannot simply take the average cash share in Clover and apply it to the settlement data. Instead, we need to model how cash usage varies with observable merchant characteristics and then use that model to predict cash usage in the settlement data.

I.B Methodology and Results of Cash Prediction Model

We find that our two-stage approach effectively captures the variation in cash usage across merchants in the Clover data. The resulting model matches key features of the distribution of cash use in the Clover data and is consistent with several pieces of external evidence.

I.B.1 Model Specification and Estimation Results

We estimate both parts of the cash extrapolation model on the Clover sample. The key predictors in both stages include the average ticket size of card transactions, merchant-level share of online transactions, firm size, sector fixed effects, and local demographic controls. The strongest predictors of cash usage are average ticket size, e-commerce participation, firm size, and the share of card transactions on debit cards. The model also includes demographic controls (population density, median household income, education, age, race/ethnicity) and industry fixed effects.

Ticket Size. Higher ticket sizes are strongly associated with lower cash usage on both margins. Figures 1a and 2a illustrate this univariate relationship in the Clover data: cash usage declines sharply as average ticket size increases.

E-commerce Status. E-commerce merchants have substantially lower cash usage, as expected. Figures 1b and 2b show the difference in cash usage on the extensive and intensive margins, respectively, between brick-and-mortar and e-commerce merchants.

Firm Size. Larger firms tend to have higher probability of accepting cash but lower conditional cash shares. Figures 1c and 2c display how the extensive and intensive margins of cash usage vary with firm size.

Debit Share. For most of the sample, higher debit share is associated with higher cash usage on both margins. This is intuitive given that debit cards are a closer substitute for cash than they are for credit cards. This pattern attenuates at very high debit shares (above roughly 80%). This non-linear relationship is captured by including both debit share and its square in the regression. Figures 1d and 2d illustrate this relationship.

I.B.2 Model Performance

The model does an adequate job of predicting both the extensive and intensive margin of cash use. The extensive margin model achieves an AUC of 0.82, indicating good discriminative ability in predicting which merchants have more than 2% of their transactions on cash.

The intensive margin model achieves a correlation of 0.52 between predicted and actual log conditional cash shares. Figure 3 shows the fit of the predicted conditional cash share distribution to the actual distribution in the Clover data, validating that the model captures the relevant conditional means of cash usage.

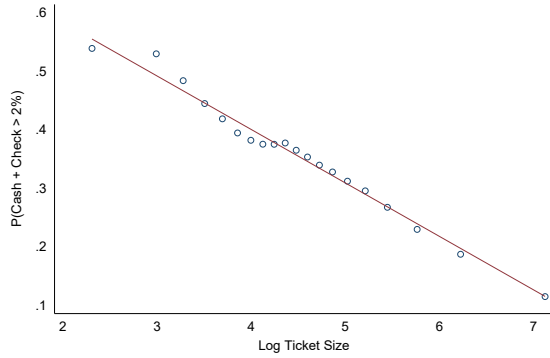
I.B.3 Aggregate Validation against Nilson Reports and Homescan Data

Our sufficient-statistics approach relies on accurately estimating the second moments of payment shares across merchants. Therefore, it is important that our cash extrapolation model accurately captures the overall level of cash usage in the economy and its variation across merchants. After we fit the cash prediction model on the Clover data, we apply it to the settlement data to predict cash usage across the settlement data. When extrapolating to the settlement data, we cap the log card sales variable at the 99th percentile of the Clover distribution to avoid extreme extrapolations along the firm size dimension.

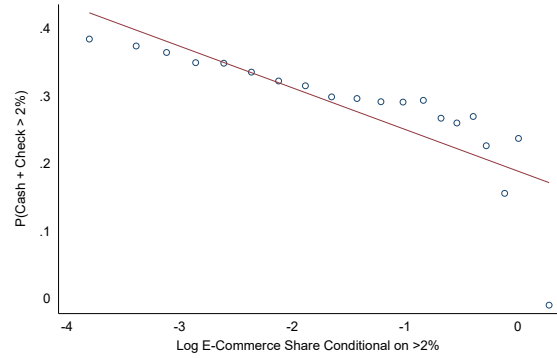
We find that our model accurately recovers both the overall level of cash usage and its variation across merchants in the settlement data. According to the 2022 Nilson Report, the average use of cash in retail transactions is around 10% (?). Our model predicts an average cash share of 10% across the settlement data, closely matching the Nilson benchmark. Appendix E

of ? reports the distribution of cash share across Homescan merchants, and finds a standard deviation of approximately 8-10 percentage points. Our model predicts a standard deviation of 9 percentage points across the settlement data, again closely matching the Homescan benchmark.

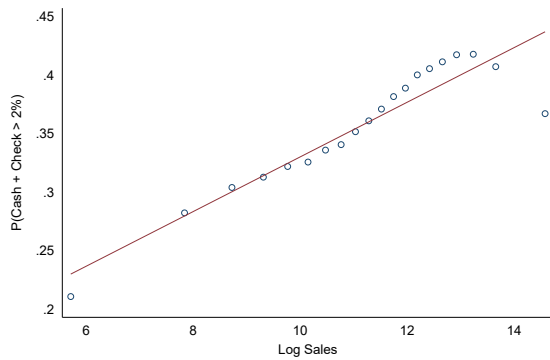
Figure 1: Probability of Positive Cash by Merchant Characteristics (Extensive Margin)



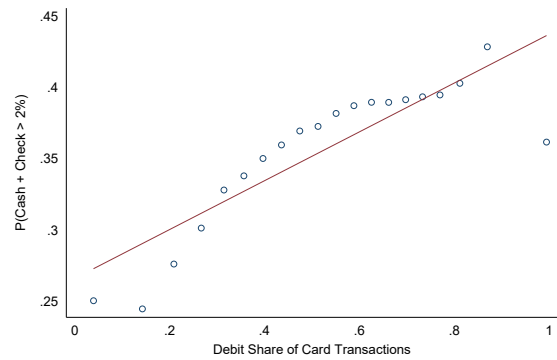
(a) By Average Ticket Size



(b) By E-commerce Status



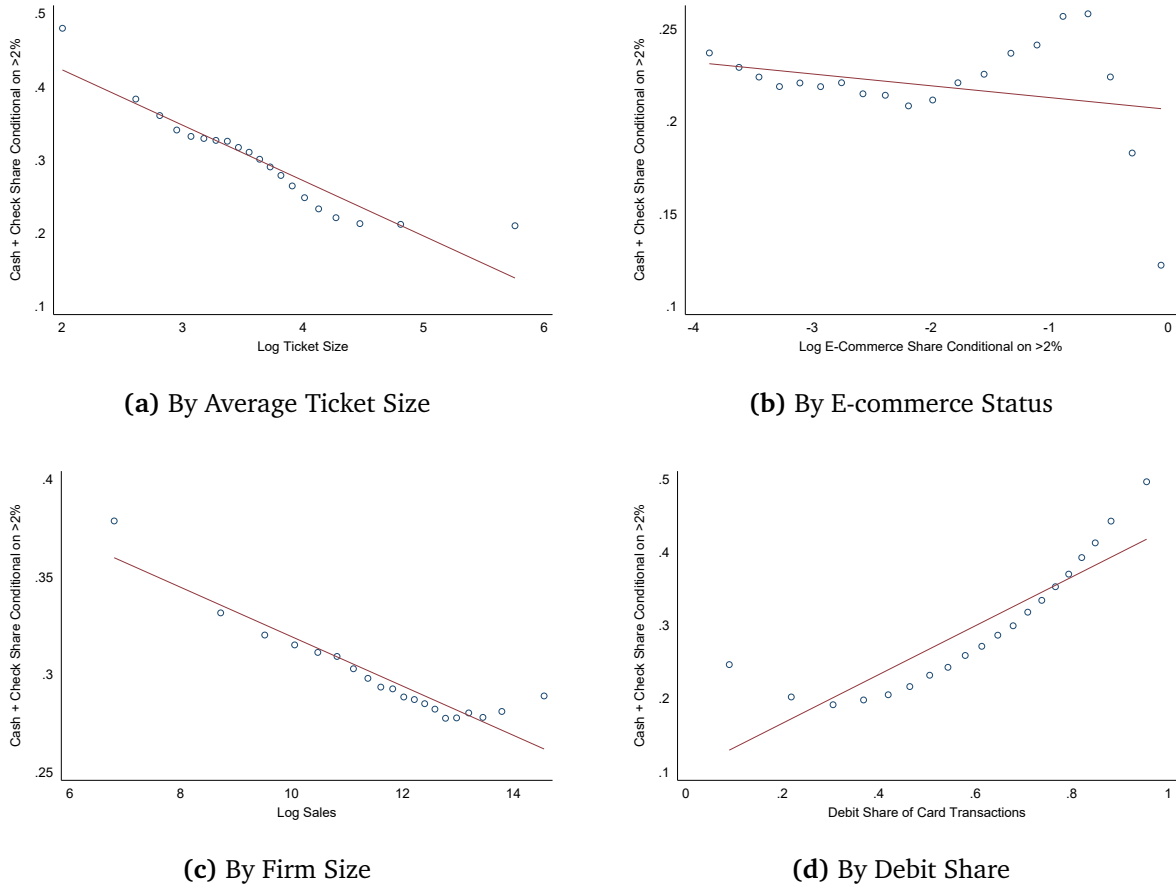
(c) By Firm Size



(d) By Debit Share

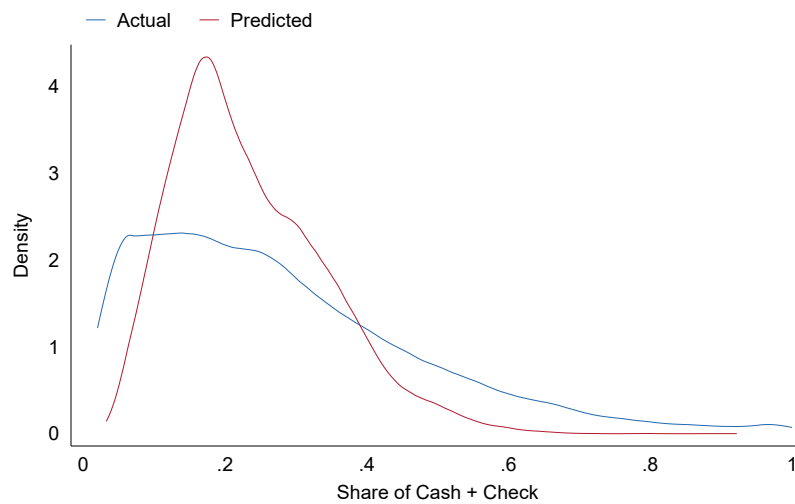
Notes: Figure shows the probability that a merchant has more than 2% of transactions in cash, by merchant characteristics in the Clover data. Panel (a): Higher ticket sizes are associated with lower probability of cash usage. Panel (b): E-commerce merchants are less likely to have positive cash usage. Panel (c): Larger firms are more likely to accept cash. Panel (d): Higher debit share is generally associated with higher probability of cash usage.

Figure 2: Conditional Cash Share by Merchant Characteristics (Intensive Margin)



Notes: Figure shows the conditional cash share among merchants with more than 2% cash usage in the Clover data, by merchant characteristics. Panel (a): Higher ticket sizes are associated with lower cash usage. Panel (b): E-commerce merchants have substantially lower cash usage than brick-and-mortar merchants. Panel (c): Larger firms tend to have lower conditional cash shares. Panel (d): Higher debit share is generally associated with higher cash usage.

Figure 3: Conditional Cash Share: Model Fit vs. Actual Distribution



Notes: Figure compares the predicted conditional cash share distribution from the model to the actual distribution in the Clover data. The figure validates that the model captures the relevant conditional means of cash usage across merchants.